
DA6701 : Data Science and AI for Finance

Assignment V

Goal-Based Portfolio Optimisation

Name	Roll Number
Naveen US	DA25M020
Rishabh Mishra	DA25M025
Tanmay Gawande	DA25M030
Nitesh Kumar Shah	ID25M806

Goal-Based Portfolio Optimisation – Report

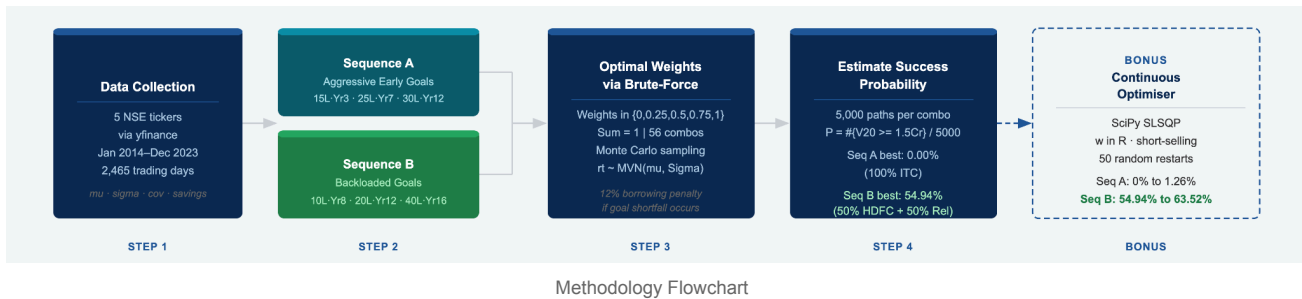
DA6701 – Assignment V | 5-Stock NSE Universe | 20-Year Horizon | Monte Carlo: 5,000 paths | Brute-Force + SciPy SLSQP

0.00%	11.00%	25.48%	54.94%	18.55%	21.12%
Seq A Success	Seq A Return	Seq A Vol	Seq B Success	Seq B Return	Seq B Vol

1 | Problem Setup & Methodology

This report presents the results of a goal-based portfolio optimisation exercise. The objective is to find a static portfolio allocation across five NSE stocks (**TCS, HDFC Bank, Reliance, Sun Pharma, ITC**) that maximises the probability of achieving a terminal retirement goal of **₹1.5 Crore at Year 20**, subject to intermediate goal withdrawals and a **12% borrowing penalty** for shortfalls.

The client contributes **₹20,000/month (₹2,40,000/year)** growing at **4% annually**, totalling approximately ₹71.5 Lakhs over 20 years. Two goal sequences are evaluated: **Sequence A with aggressive early goals** (₹15L at Year 3, ₹25L at Year 7, ₹30L at Year 12) and **Sequence B with backloaded goals** (₹10L at Year 8, ₹20L at Year 12, ₹40L at Year 16). Portfolio weights are drawn from the discrete set $\{0, 0.25, 0.50, 0.75, 1.00\}$ summing to **1.0**, yielding **56** valid combinations. Each combination is evaluated via **5,000-path Monte Carlo** simulation using annualised return and covariance parameters estimated from 10 years of daily log returns (Jan 2014 – Dec 2023).

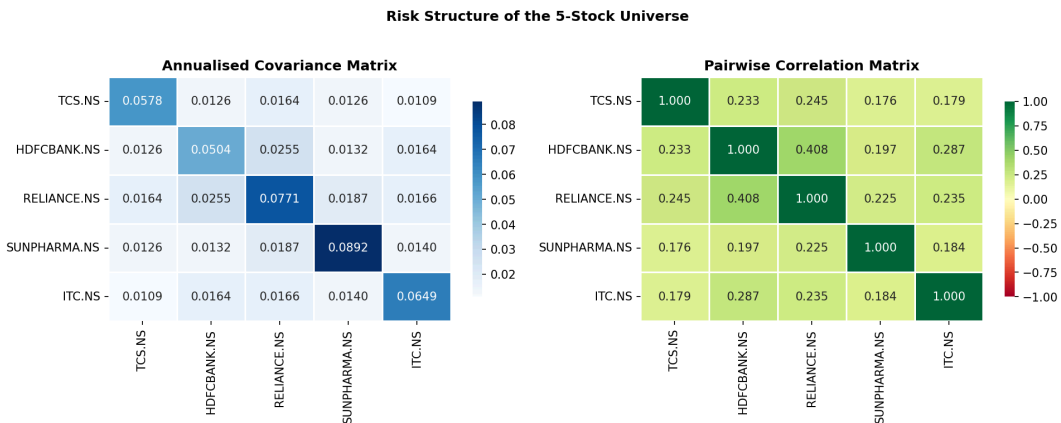


2 | Data & Annualised Statistics

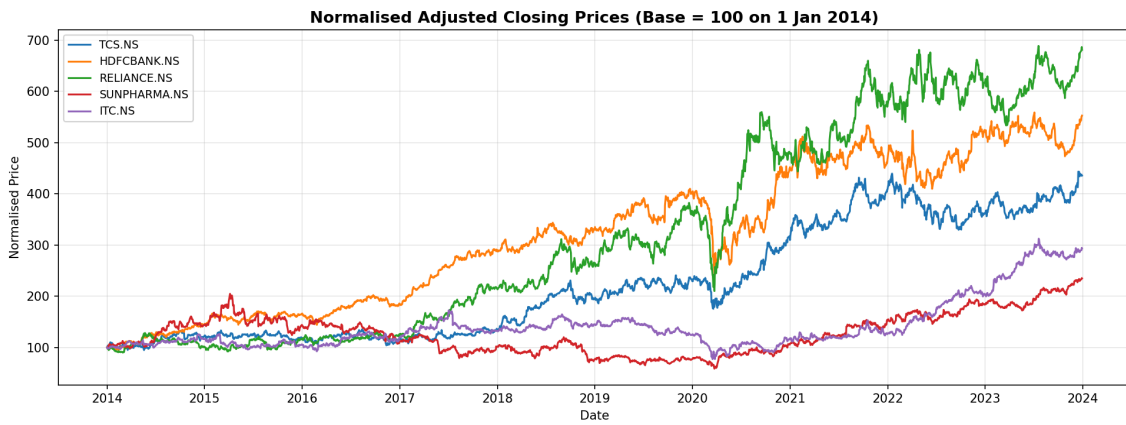
Daily adjusted closing prices for **2,465 trading days** were downloaded via `yfinance`. **Log returns** were computed and **annualised** (x252) to obtain **expected returns** and the **covariance matrix**. The table below summarises the per-stock statistics.

Ticker	Ann. Return	Ann. Std Dev
TCS.NS	15.05%	24.04%
HDFCBANK.NS	17.49%	22.46%
RELIANCE.NS	19.62%	27.77%
SUNPHARMA.NS	8.73%	29.86%
ITC.NS	11.00%	25.48%

Reliance has the **highest annualised return (19.62%)** but also elevated volatility. **Sun Pharma** shows the **lowest return (8.73%)** with the **highest volatility (29.86%)**, making it the least efficient on a standalone basis. All pairwise correlations in the covariance matrix are moderate, providing diversification potential.



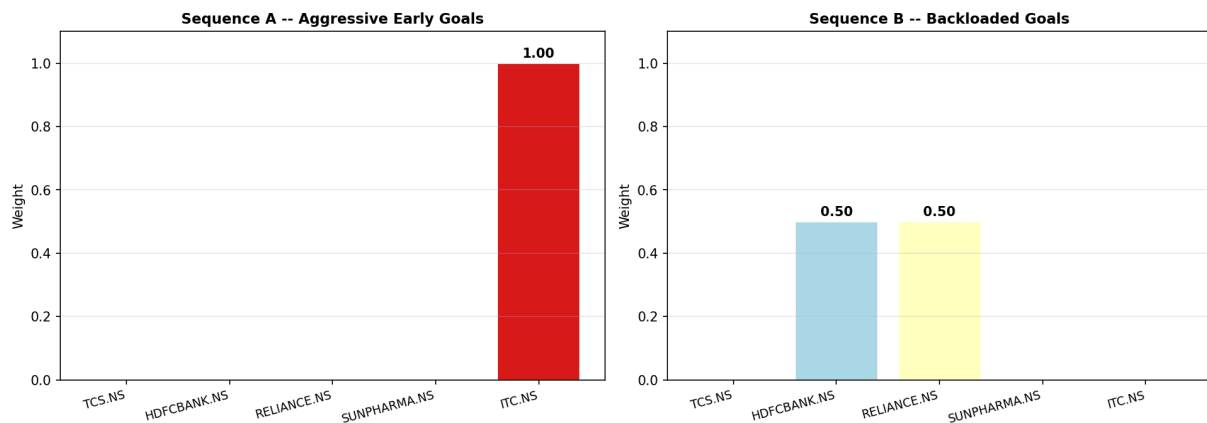
Covariance & Correlation Heatmaps



3 | Optimal Portfolio Weights – Brute-Force Results

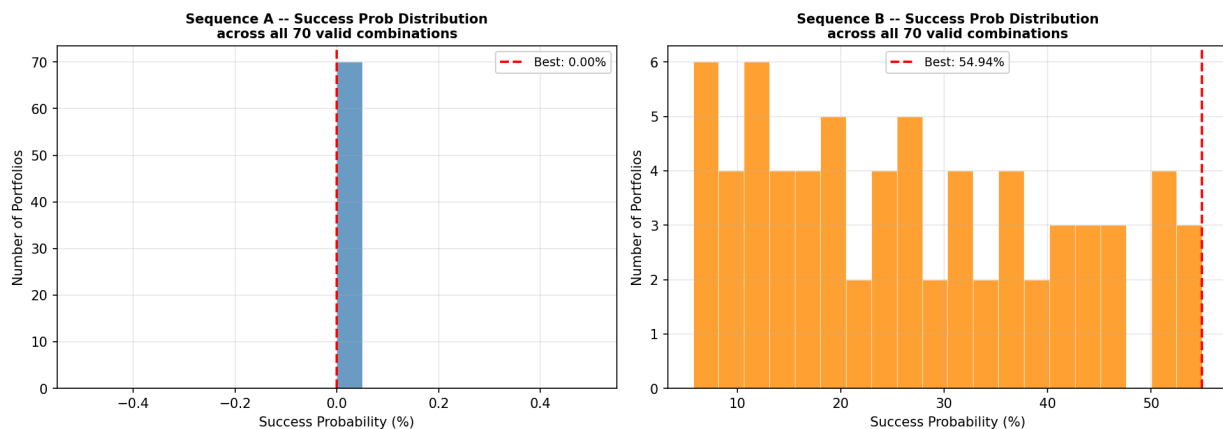
Ticker	Seq A Weight	Seq B Weight
TCS.NS	0.00	0.00
HDFCBANK.NS	0.00	0.50
RELIANCE.NS	0.00	0.50
SUNPHARMA.NS	0.00	0.00
ITC.NS	1.00	0.00
Success Probability	0.00%	54.94%

Optimal Portfolio Weights: Sequence A vs Sequence B



Optimal Weight Bar Charts – Sequence A vs Sequence B

Brute-Force Search Results: All 56 Discrete Portfolios

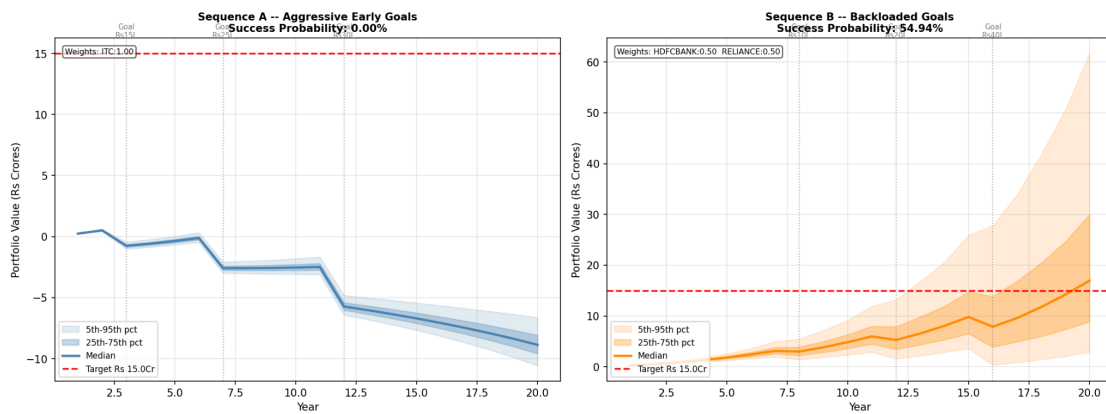


Success Probability Distributions

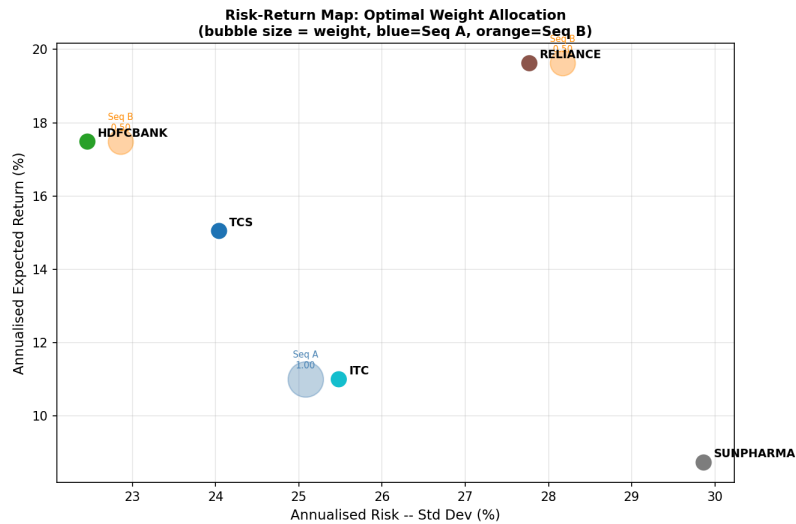
Full Results Summary

Metric	Sequence A	Sequence B
TCS.NS weight	0.00	0.00
HDFCBANK.NS weight	0.00	0.50
RELIANCE.NS weight	0.00	0.50
SUNPHARMA.NS weight	0.00	0.00
ITC.NS weight	1.00	0.00
Portfolio Exp. Return (ann.)	11.00%	18.55%
Portfolio Std Dev (ann.)	25.48%	21.12%
Success Probability (Optimal)	0.00%	54.94%
Success Probability (Equal Weight)	0.00%	24.34%
Improvement over Equal Weight	+0.00%	+30.60%

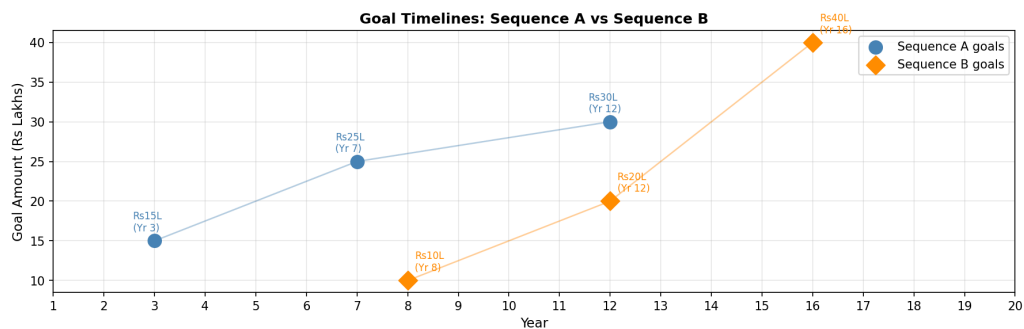
Optimal Portfolio -- Percentile Fan Charts (5,000 Paths)



Year-by-Year Percentile Fan Charts (Optimal Portfolios)



Risk-Return Scatter Plot



Goal Timelines: Sequence A vs Sequence B

4 | Results Discussion – Why Different Goals Produce Different Portfolios

Sequence A (100% ITC, 0.00% success): This sequence demands ₹15L by Year 3 and ₹25L by Year 7. The portfolio starts at zero and accumulates **only ~₹7.5L** in savings over the first three years- meaning even with strong returns, meeting the Year 3 goal is *nearly impossible*. Almost every simulated path **borrow**s at Year 3, and that loan compounds at **12% for the remaining 17 years**. By Year 7, the portfolio is already **underwater** when the **second ₹25L goal** hits, triggering more borrowing. **The debt spiral is unrecoverable.**

The **optimiser picks ITC** (the **lowest-volatility** stock at **25.48%**) not because it expects to succeed, but because it is **minimising damage**. With **0% success** across all **56 combinations**, the optimiser defaults to the **least volatile option** - the one that produces the least negative final values. A higher-return, higher-volatility pick like Reliance would just amplify the losses on the downside without meaningfully helping on the upside, since no combination of weights can overcome the structural impossibility of meeting ₹15L from ₹7.5L of savings in 3 years.

Sequence B (50% HDFCBANK + 50% RELIANCE, 54.94% success): The first goal is **only ₹10L at Year 8**. By then the portfolio has had **eight years of uninterrupted** compounding and **~₹21L in cumulative savings**, so the majority of paths can meet the goal *without borrowing* at all. With borrowing risk low, the optimiser is free to pursue **return maximisation**. It picks the **two highest-return stocks** - **HDFCBANK (17.5% ann. return, 22.5% std)** and **RELIANCE (19.6% return, 27.8% std)** - which together deliver an **18.55% portfolio return** at a **blended 21.12% std dev** (lower than either stock alone, thanks to imperfect correlation).

The core insight: *The goal sequence does not just change which portfolio is optimal, it changes what the optimiser is even trying to do. Sequence A turns the problem from “maximise success probability” into “minimise guaranteed failure,” while Sequence B keeps it a genuine optimisation where diversification and return both matter. This is the fundamental lesson of goal-based investing: “the timing and size of intermediate obligations can completely dominate the investment decision, overriding any analysis of the assets themselves.”*

5 | Bonus – Continuous SciPy Optimiser (SLSQP with Short-Selling)

The continuous optimiser replaces the discrete brute-force grid with **SciPy's SLSQP** method. The only constraint is `sum(weights) = 1.0` with no bounds on individual weights, permitting short positions of any size. A 50-restart strategy (half Dirichlet long-only starts, half unconstrained random starts) is used to navigate the non-convex, stochastic objective landscape.

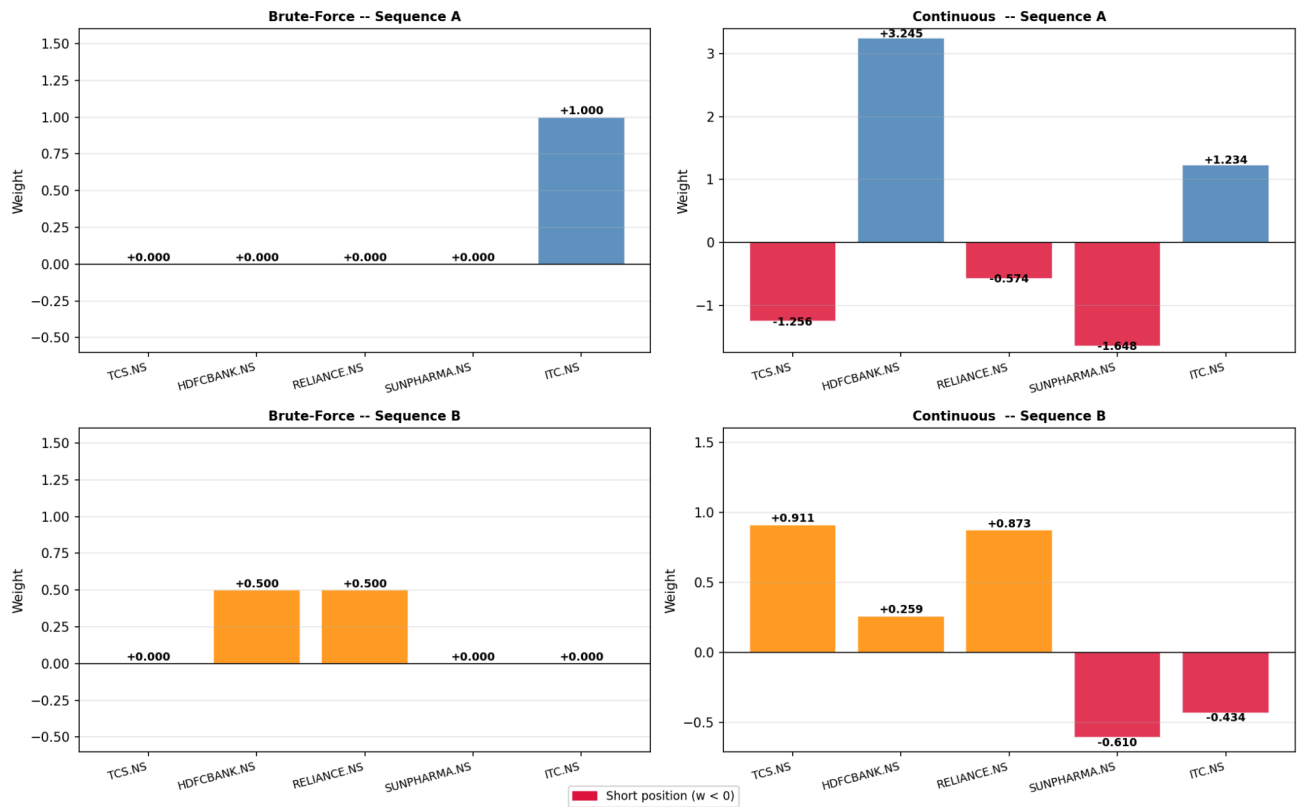
Three-Way Comparison: Equal Weight vs Brute-Force vs Continuous

	Equal Wt	BF (A)	Cont. (A)	BF (B)	Cont. (B)
TCS.NS	0.20	0.00	-1.2563	0.00	0.9114
HDFCBANK.NS	0.20	0.00	3.2449	0.50	0.2594
RELIANCE.NS	0.20	0.00	-0.5743	0.50	0.8735
SUNPHARMA.NS	0.20	0.00	-1.6480	0.00	-0.6105
ITC.NS	0.20	1.00	1.2336	0.00	-0.4338
Success Prob	0.00%	0.00%	1.26%	54.94%	63.52%
vs Equal-Wt	--	+0.00%	+1.26%	+30.60%	+39.18%
vs Brute-Force	--	--	+1.26%	--	+8.58%

Sequence A- from 0.00% to 1.26%: The continuous optimiser broke through by finding a heavily leveraged portfolio: **short TCS (-1.26)**, **short Reliance (-0.57)**, **short SunPharma (-1.65)**, and **long HDFCBANK (3.24×)** and **ITC (1.23×)**. This extreme leverage amplifies the portfolio's expected return, giving a **small fraction of paths** enough explosive early growth to clear the Year 3 and Year 7 goals. The **1.26%** success rate is still very low, but demonstrates that short-selling can extract marginal probability from an otherwise impossible problem.

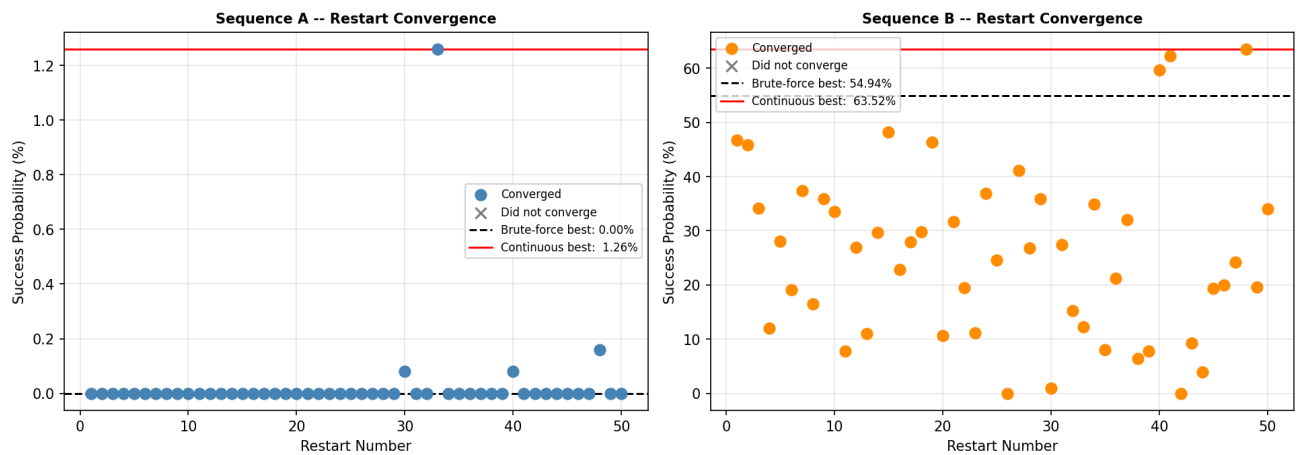
Sequence B- from 54.94% to 63.52% (+8.58%): The continuous solution keeps **heavy long** positions in **TCS (0.91)** and **Reliance (0.87)**, adds **moderate HDFCBANK (0.26)**, and funds this by **shorting SunPharma (-0.61)** and **ITC (-0.43)**, the two lowest-return stocks. This is a textbook long-short construction: borrow from low-return assets to overweight high-return ones. The **+8.58% improvement over brute-force** comes entirely from accessing weight combinations that the discrete grid cannot represent.

Portfolio Weights: Brute-Force vs Continuous Optimiser



Brute-Force vs Continuous Weight Bar Charts

Continuous Optimiser: Success Probability Across All Restarts



Restart Convergence Plots

Key takeaway: The continuous optimiser improves on brute-force in both sequences, but the nature of the improvement differs. For **Sequence A**, short-selling turns **an impossible problem into a barely possible one** (0% → 1.26%) through extreme leverage - theoretically interesting but practically risky. For **Sequence B**, it delivers **a meaningful, implementable improvement** (54.94% → 63.52%) through moderate long-short positioning. This contrast reinforces the Section 4 finding: the goal structure determines not just the optimal allocation, but the kind of strategy that can help.